Potenziale Em

$$B = \nabla \times A \qquad \nabla \times E = -\frac{\partial B}{\partial t}$$

$$\Rightarrow \nabla \times E = -\frac{\partial}{\partial t}(\nabla \times A)$$

$$\nabla \times E = -\nabla \times \frac{\partial A}{\partial t}$$

$$\nabla \times \left(E + \frac{\partial A}{\partial t}\right) = 0$$

NB $\nabla \times (\nabla V) = 0 \quad \forall V$

$= \exists\, V$ scalare :

$$E + \frac{\partial A}{\partial t} = -\nabla V$$

$$\Rightarrow E = -\frac{\partial A}{\partial t} - \nabla V$$

NB: $A' = A + \nabla \varphi$

$$\nabla \times A' = \nabla \times A + \cancel{\nabla \times \nabla \varphi} \quad \forall \varphi$$

$\exists$ una famiglia $A' = A + \varphi$:

$$\nabla \times A' = \nabla \times A$$

$$\Rightarrow B = \nabla \times A'$$

$$E = -\frac{\partial A'}{\partial t} - \nabla V'$$

$$= -\frac{\partial}{\partial t}(A + \nabla \varphi) - \nabla V'$$

$$= -\frac{\partial A}{\partial t} - \nabla \frac{\partial \varphi}{\partial t} - \nabla V'$$

$$-\frac{\partial A}{\partial t} - \nabla\left(\frac{\partial \varphi}{\partial t} + V'\right) = -\frac{\partial A}{\partial t} - \nabla V$$

$$V = \frac{\partial \varphi}{\partial t} + V'$$

$\Rightarrow$ GAUGE CONDITIONS

$$V' = V - \frac{\partial \varphi}{\partial t}$$

$$A' = A + \nabla \varphi$$

I potenziali che soddisfano tali condizioni generano gli stessi campi EM

CONDIZIONE di COULOMB/MAXWELL

$$\nabla \cdot A = 0$$

$$\nabla \times A = B = \nabla \times A'$$

$$\nabla \cdot A' = 0$$

$$\Rightarrow \nabla(A + \nabla \varphi) = 0$$

$$\Rightarrow \nabla \cdot \nabla \varphi = 0$$

$$\Rightarrow \nabla^2 \varphi = 0$$

$$\Rightarrow \varphi = \text{cost}$$

$$\Rightarrow A' = A$$

$$V' = V$$

$$\nabla \cdot E = \nabla\left(-\frac{\partial A}{\partial t} - \nabla V\right) = \frac{\rho}{\varepsilon_0}$$

$$-\frac{\partial \nabla \cdot A}{\partial t} - \nabla^2 V = \frac{\rho}{\varepsilon_0}$$

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}$$

$$V = \frac{1}{4\pi\varepsilon_0} \iiint_T \frac{\rho}{\varepsilon_0}\, d\tau$$

$$\nabla \times B = \mu_0\left(J + \varepsilon_0 \frac{\partial E}{\partial t}\right)$$

$$\nabla \times (\nabla \times A) = \mu_0\left(J + \varepsilon_0 \frac{\partial E}{\partial t}\right)$$

$$\nabla(\nabla \cdot A) - \nabla^2 A = \mu_0\left(J + \varepsilon_0 \frac{\partial E}{\partial t}\right)$$

$$A = \frac{\mu_0}{4\pi} \iiint_T \frac{J + \varepsilon_0 \frac{\partial E}{\partial t}}{r}\, d\tau$$

CONDIZIONE di LORENTZ

$$\boxed{\nabla \cdot A + \varepsilon_0 \mu_0 \frac{\partial V}{\partial t} = 0}$$

$$\nabla(A + \nabla \varphi) + \varepsilon_0 \mu_0 \frac{\partial}{\partial t}\left(V - \frac{\partial \varphi}{\partial t}\right) = 0$$

$$\nabla \cdot A + \nabla^2 \varphi + \varepsilon_0 \mu_0 \frac{\partial V}{\partial t} - \varepsilon_0 \mu_0 \frac{\partial^2 \varphi}{\partial t^2} = 0$$

$$\Rightarrow \nabla^2 \varphi - \varepsilon_0 \mu_0 \frac{\partial^2 \varphi}{\partial t^2} = 0$$

φ risolve eq onde

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\nabla \times B = \mu_0\left(J + \varepsilon_0 \frac{\partial E}{\partial t}\right)$$

$$E = -\frac{\partial A}{\partial t} - \nabla V$$

$$B = \nabla \times A$$

$$\nabla \times B = \nabla(\nabla \cdot A) - \nabla^2 A$$

$$\nabla(\nabla \cdot A) - \nabla^2 A = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t}$$

$$\nabla(\nabla \cdot A) - \nabla^2 A = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial}{\partial t}\left(-\frac{\partial A}{\partial t} - \nabla V\right)$$

$$-\nabla \cancel{\mu_0 \varepsilon_0 \frac{\partial V}{\partial t}} - \nabla^2 A = \mu_0 J - \mu_0 \varepsilon_0 \frac{\partial^2 A}{\partial t^2} - \cancel{\mu_0 \varepsilon_0 \nabla \frac{\partial V}{\partial t}}$$

$$\nabla^2 A - \mu_0 \varepsilon_0 \frac{\partial^2 A}{\partial t^2} = -\mu_0 J$$

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\nabla\left(-\frac{\partial A}{\partial t} - \nabla V\right) = \frac{\rho}{\varepsilon_0}$$

$$-\frac{\partial \nabla \cdot A}{\partial t} - \nabla^2 V = \frac{\rho}{\varepsilon_0}$$

$$\varepsilon_0 \mu_0 \frac{\partial^2 V}{\partial t^2} - \nabla^2 V = \frac{\rho}{\varepsilon_0}$$

$$\nabla^2 V - \mu_0 \varepsilon_0 \frac{\partial^2 V}{\partial t^2} = -\rho / \varepsilon_0$$

Se $\rho = 0$

$$V' = V - \frac{\partial \varphi}{\partial t}$$

$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = 0$$

Se scelgo

$$V = -\frac{\partial \varphi}{\partial t} \Rightarrow V = 0$$

$$B = \nabla \times A \qquad E = -\frac{\partial A}{\partial t}$$

CAMPO EM descritto solo da A